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Solar transparent and thermally insulated silica aerogel for efficiency improvement of photovoltaic/thermal collectors

Lijun Wu, Bin Zhao*, Datong Gao, Dongsheng Jiao, Maobin Hu and Gang Pei*

Abstract

Photovoltaic/thermal (PV/T) utilization has been regarded as a promising technique to efficiently harvest solar energy, but its thermal efficiency highly degrades in cold seasons because of remarkable heat loss. Although various methods, such as using air or vacuum gap, have been used to reduce heat loss of the PV/T, heat radiative loss still exists. In addition, unlike selective solar absorbers, the current PV/T absorber behaves like an infrared blackbody, showing great radiative heat loss. To overcome this drawback, a novel aerogel PV/T (referred to as “A-PV/T” hereinafter) collector based on solar transparent and thermally insulated silica aerogel is proposed, which can reduce the heat loss from both the non-radiative and radiative heat transfer modes. Experimental testing demonstrates that the thermal efficiency improvement of 25.1%–348% can be achieved for PV/T within the collecting temperature range of 35–70 °C when silica aerogel is introduced, indicating a significant efficiency enhancement. Compared with traditional PV/T (referred to as “T-PV/T” hereinafter) collector, the stagnation temperatures of the A-PV/T collector are 96.7 °C and 103.1 °C in outdoor and indoor environments, which are 27.4 °C and 25.8 °C greater, respectively, indicating a heat loss suppression of the aerogel. Moreover, simulation reveals that useful heat can hardly be provided by the T-PV/T collector in cold seasons, but the A-PV/T still exists a high solar thermal performance, showing good seasonal and regional applicability.

Keywords Photovoltaic/thermal (PV/T), Radiative heat loss, Silica aerogel, Solar energy, Radiative transfer

1 Introduction

Solar energy is a plentiful green energy resource and can alleviate society's dependence on fossil fuels [1–4]. Photovoltaic/thermal (i.e., PV/T) utilization combines photovoltaic and photothermal processes to generate clean electricity and heat in one device, by converting part of sunlight into electricity and the rest of solar irradiance into heat that is collected by working

fluids as the heat source for other purposes [5–9], which exhibits a significant performance improvement in comparison to the standalone photovoltaic module. Reviewing the published papers, PV/T technology has already been used in various applications to establish near-zero energy structures. Souliotis et al. [10] constructed a life cycle assessment analysis of combining PV/T collectors and building roofs to generate more energy, which contrasts vividly with standalone PV and solar thermal panels. Das et al. [11] proposed a new PV/T collector that has a rectangular spiral tube and a phase change materials-based composite, which has a high thermal efficiency of 66.6% and improves electrical power by 18.4%. Liang et al. [12] theoretically evaluated the influence of the PV/T collector installed on the wall on buildings. PV/T collectors not

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only supply hot water and electricity for buildings but also act as a wind barrier to suppress building thermal loss. The simulation result shows that the heat loss of the external wall is suppressed by approximately 40%, which increases the water supply temperature to 40 °C for buildings. Agrawal and Tiwari [13] optimized PV/T systems under cold weather conditions, and the system after optimization can provide 15,209 kWh of electrical energy and 18,062 kWh of thermal energy in one year with a constant air mass rate.

In fact, in the cold season, PV/T collectors show unsatisfactory thermal performance. The main drawback is that the thermal performance dramatically deteriorates as the ambient temperature decreases. The poor thermal performance makes it impossible to provide useful heat for buildings in cold conditions (e.g., cold winter and locations) without an auxiliary heat source [14]. In addition, compared to the solar collector with a spectral selective absorber, the PV/T collector presents a lower capacity for heat generation because the PV/T absorber is generally encapsulated by the optical transparent Tedlar/Polyester/Tedlar (hereinafter referred to as “TPT”) with a high infrared emissivity of approximately 0.9 [15], and the air gap is transparent to infrared radiation, so the radiative heat loss is enormous and thus the thermal efficiency is at a low level.

To solve this issue, many contributions have been made to the topic of radiation heat loss suppression. An aluminum-doped ZnO coating on a glass surface is developed [16], which shows a low infrared emissivity (~ 0.3) and a high solar transmittance (~ 0.79), and a thermal efficiency enhancement of 45% was obtained. Lämmle et al. [17] designed suitable spectrally selective coatings by applying an optical coating consisting of aluminum-doped ZnO and reflective metal film to obtain a solar transmittance (79%) while presenting an ultralow emissivity of 0.13 at 373 K. Our group [18] also reveals that the importance of emissivity of PV cells for PV/T hybrid utilization. Simulation results show that a significant efficiency enhancement with the longwave emissivity dropping from 0.95 to 0.05 is obtained. The above consequences manifest that reducing the longwave emissivity can make a significant thermal efficiency improvement. The above experiments and simulations were carried out by optimizing and regulating the spectral selectivity of the PV/T absorber. However, there are few experimentally verified works, and the relevant coatings exhibit large optical loss so that the PV cell cannot generate electricity efficiently. Therefore, simultaneously suppressing radiative heat loss based on the spectral selective method and maintaining the high optical efficiency for electrical generation is a potential method for PV/T technology to improve harvesting efficiency.

Based on the thermal analysis, introducing a low thermal conductive radiation shield into the gap between PV/T absorbers and glass envelopes is a useful method to reduce heat loss. In our previous studies [19], an aerogel was introduced on the top PV/T absorber surface for heat loss suppression. Aerogels are good materials that can be designed to exhibit ultralow density [20–23], ultrahigh porosity [24, 25], ultralow thermal conductivity, and ultrahigh-temperature resistance. The greatest advantage is that the silica aerogel can work as an optical filter; it presents opaque to infrared radiation but is transparent to sunlight [19]. Moreover, the aerogel has an ultralow thermal conductivity, which can be designed to be smaller than that of ambient air [26]. The spectral properties and physical properties of the aerogel make it extremely effective in heat loss suppression. Recently, the transparent silica aerogel attracts the researchers' attention due to its excellent insulation performance. The silica aerogel prevents heat loss while having high sunlight transmit capacity which is mainly determined by its transparency to sunlight, opaque in the IR, and extremely low effective thermal conductivity. Zhao et al. [23] constructed a device that reached over 473 K under realistic weather without the demand of the optical concentrator by combining the silica aerogel with the flat-plate solar receiver. Moreover, Zhao et al. [27] obtained the silica aerogel with an extremely high solar transmittance (95%) by optimizing its microstructure to reduce the scattering of the aerogel. On this basis, they demonstrated a solar thermal device based on silica aerogel, and the device can generate steam at 100 °C with an efficiency of 56% under the 700 W/m² solar irradiation. Qiu et al. [28] proposed a novel parabolic trough receiver insulated by silica aerogel, reporting the efficiency of the parabolic trough receiver can be improved by 0.32–5.04%. In these applications, aerogel is generally introduced to the solar thermal absorbers, the knowledge gap of aerogel for PV/T applications is being explored. Although few works focus on integrating aerogel with PV/T collectors, such as the aerogel-based PVT-TEG system [29], a detailed radiative heat transfer mechanism is not considered and the existing aerogel-based solar absorbers are designed to improve working temperatures, while aerogel-based PV/T in this work is mainly to improve efficiency in cold conditions and extend the operating temperature region. Compared to the existing heat suppression methods (e.g., vacuum method) of PV/T collector, aerogel can significantly reduce the heat loss of the PV/T absorber from both the heat conduction and heat radiation suppressions, which is a new try for highly efficient PV/T hybrid utilization.

Motivated by the above analysis, this work aims to experimentally explore the performance enhancement

after using an aerogel as a radiation shield for PV/T hybrid utilization. Experimental demonstration is performed to explore the performance improvement of the aerogel PV/T (referred to as “A-PV/T” hereinafter) collector. A indoor experiment is carried out based on the solar simulator under unconcentrated solar flux, and the stagnation temperature, thermal and electrical efficiency are tested, compared, and analyzed. In addition, an outdoor experimental platform is constructed to reveal the A-PV/T’s performance in realistic weather conditions. Finally, a validated numerical model is applied to predict the A-PV/T performance in different regions.

2 Description of the experimental setup

2.1 Transparent silica aerogel

The 12.5-mm-thick silica aerogel used in this work is solar transparent (Fig. 1(a)) and the transmittance (Fig. 1(b)) of the aerogel was measured based on a spectrophotometer (SolidSpec-3700) from 0.3 to 2.5 μm and an FT-IR spectrometer (Nicolet iS10) from 2.5 to 20 μm . The aerogel presents a high transmittance to solar irradiance with an AM 1.5 weighted transmittance of up to 87%. In addition, the aerogel has ultralow transmittance in the infrared band, which greatly reduces the radiative heat loss. Moreover, the thermal conductivity of the aerogel was tested by a laser thermal conductivity testing instrument (NETZSCH LFA457). The aerogel exhibits an ultralow thermal conductivity of approximately $0.0254 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, which is

lower than ambient air ($\sim 0.026 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ [30]), and this property mainly benefits from the highly porous structure [31]. Besides, the silica aerogel is lightweight with a density of about $0.09 \text{ g}\cdot\text{cm}^{-3}$ and this is also contributed by the highly porous structure.

2.2 The prototype of the A-PV/T

Figure 2(a) shows a photo and schematic diagram of the A-PV/T collector which consists of a transparent silica aerogel, an encapsulated PV/T absorber, and an insulation material. The dimension of the collector is $170 \text{ mm} \times 170 \text{ mm} \times 55 \text{ mm}$, and the dimension of the PV/T absorber and aerogel is $50 \text{ mm} \times 50 \text{ mm}$. During the experiment, a T-PV/T is set up for reference (Fig. 2(b)), and the difference between them is only the transparent silica aerogel in Fig. 2(a) substituted with the glass cover. Notably, additional protective layers, such as glass, will be optionally added on top of silica to protect the aerogel depending on its mechanical strength. Here, silica aerogel exposes to the sky directly. The amorphous silicon (a-Si) solar cell is employed for the PV/T absorber because of its low-temperature coefficient [32]. As shown in Fig. 3, the AM 1.5 solar spectrum weighted absorptivity of the cell is 0.77. The a-Si cell is fixed in an aperture, and the silica aerogel is attached to the top of the a-Si cell. The insulation material is applied to decrease the heat transfer between the absorber and ambient air.

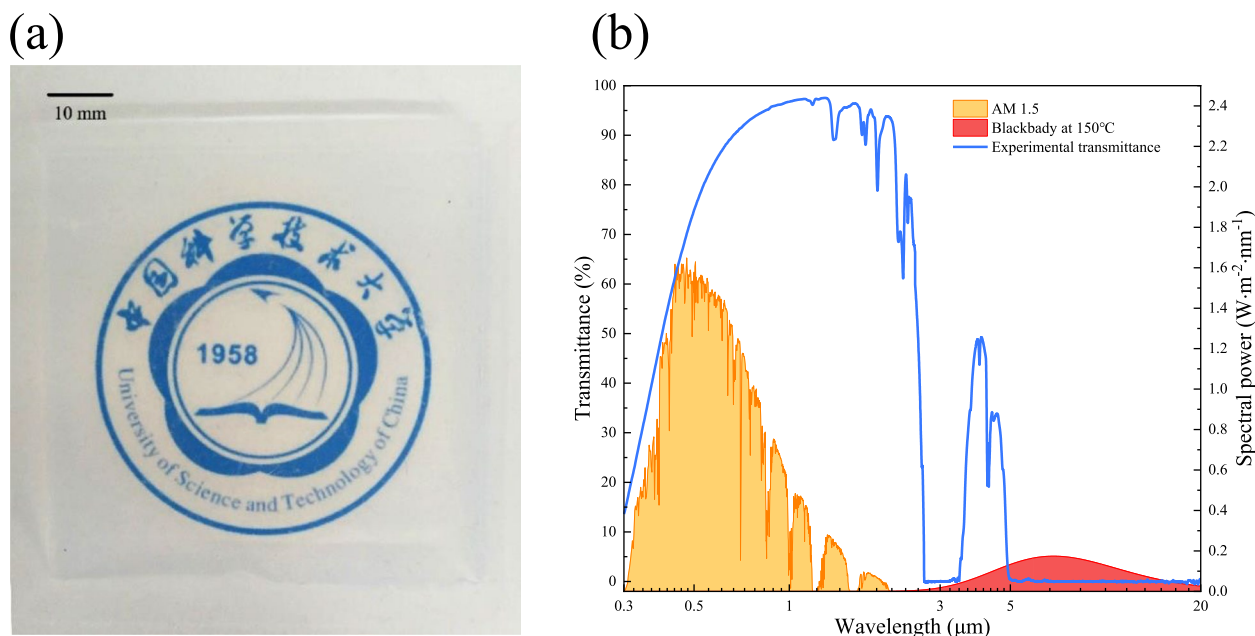


Fig. 1 **a** The image of 12.5-mm-thick transparent silica aerogel. **b** Measured transmittance curve of the aerogel ranges within 0.3~20 μm

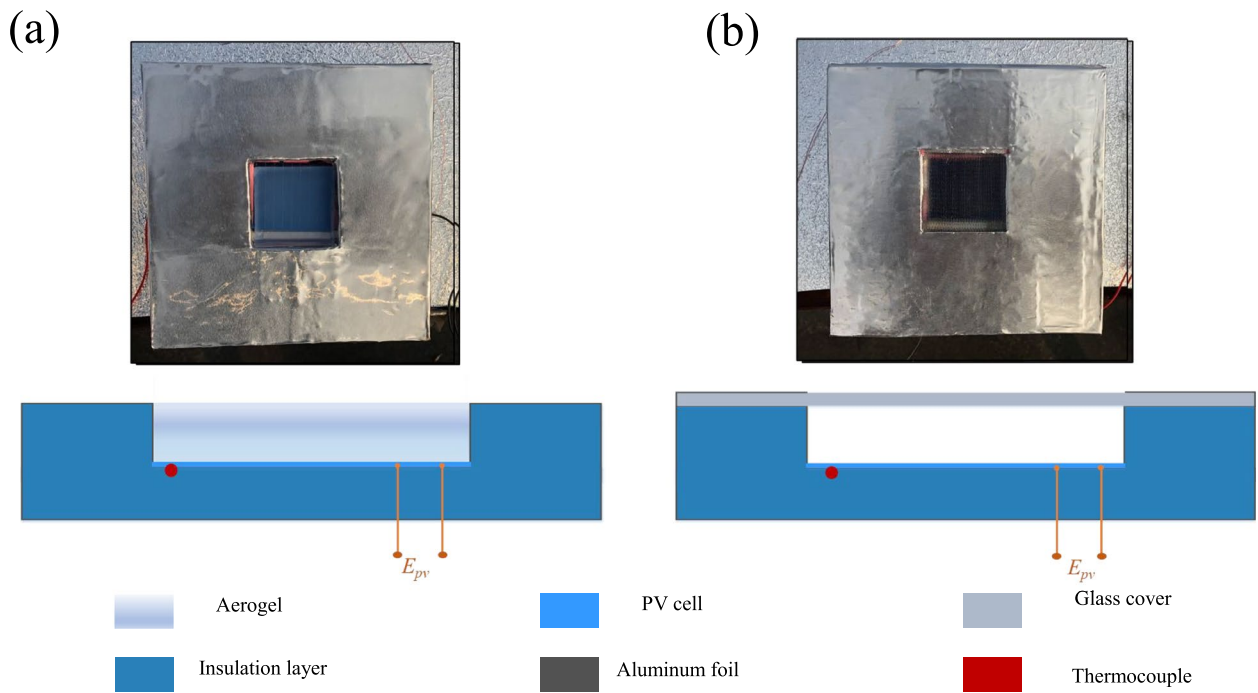


Fig. 2 Photo and schematic of the (a) A-PV/T and (b) T-PV/T

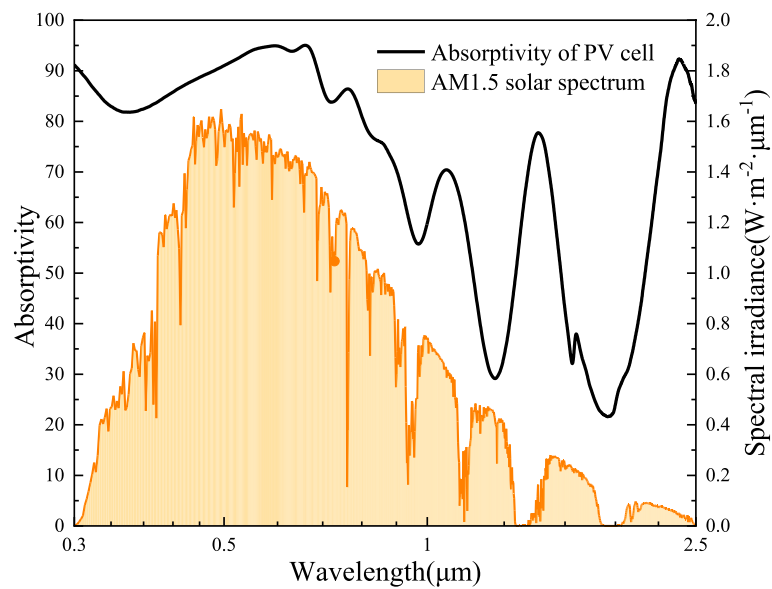


Fig. 3 Measured solar absorptivity curve of the encapsulated cell

2.3 Efficiency and stagnation temperature measurement

Thermal/electrical efficiency and stagnation temperature are important indicators for the performance evaluation of the PV/T utilization. A testing system is designed and established to measure the thermal efficiency of the fabricated prototype and the schematic is shown in Fig. 4. A

Peltier device with a PID control device is used to provide heating/cooling power for the a-Si cell to control temperature. A heat flux sensor is attached between the a-Si cell and the Peltier device to measure the input and output heat flux of the a-Si cell. To control the solar power under relatively stable conditions, a solar simulator (Oriel Sol3A

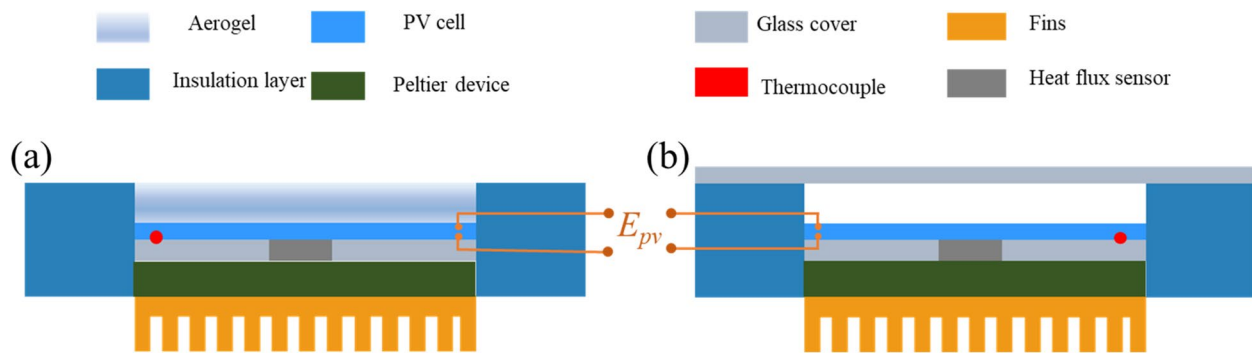


Fig. 4 Schematic of the A-PV/T and T-PV/T collectors for efficiency testing

Model 90943A, as shown in Fig. 5) is used for thermal efficiency testing. When the system is stabilized, the heat flux from solar irradiance (downward heat flux) equals the heat flux supplied by the Peltier device, and the value of the flux is monitored by the flux sensor. Thus, thermal efficiency is calculated by:

$$\eta_{th} = \frac{P \cdot V_{hf}/S_{hf}}{I_{sol}} \quad (1)$$

where P is a heat flux correction factor, which is used to eliminate the effect of the conductivity difference between the glass and heat flux sensors. S_{hf} and V_{hf} are the sensitivity and voltage output of the heat flux sensor, respectively.

To measure the electrical efficiency, I-V testing is performed under the solar simulator. Photo and schematic of the I-V testing system are displayed in Fig. 5. To obtain the electrical efficiency under various temperatures, a heater is attached to the backside of the cell, and the temperature control uncertainty is about ± 0.5 °C.

The stagnation temperature is tested under steady-state conditions. A T-type thermocouple is fixed between the a-Si cell and insulation layer to measure the transient temperature with an uncertainty of ± 0.5 °C. To comprehensively test the stagnation temperature of the two PV/T collectors, two conditions, the indoor solar simulator condition (Fig. 5(a)) and the outdoor weather condition (Fig. 6), are selected. In the outdoor condition (Fig. 6), two PV/T collectors are fixed on mechanical support and arranged toward the south direction with a slope angle of approximately 30° to obtain maximum solar irradiance. In addition, the solar irradiance reaching the collector's surface is monitored by a solar pyranometer (TBQ-2) with an uncertainty of $\pm 2\%$. The experimental data are saved by a data acquisition (HIOKI LR8402-21).

Due to the accuracy of the monitors in experiments, a general uncertainty analysis of the testing results is conducted, including thermal efficiency, stagnation temperature, and electrical efficiency. The root sum-of-the-squares method is selected for uncertainty analysis, which can be presented by:

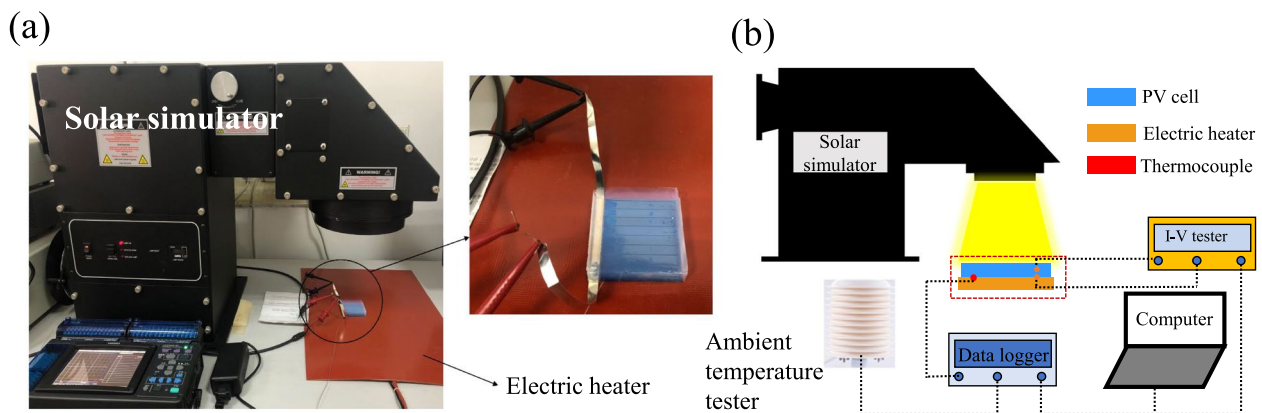


Fig. 5 Photo and schematic for measuring electrical efficiency

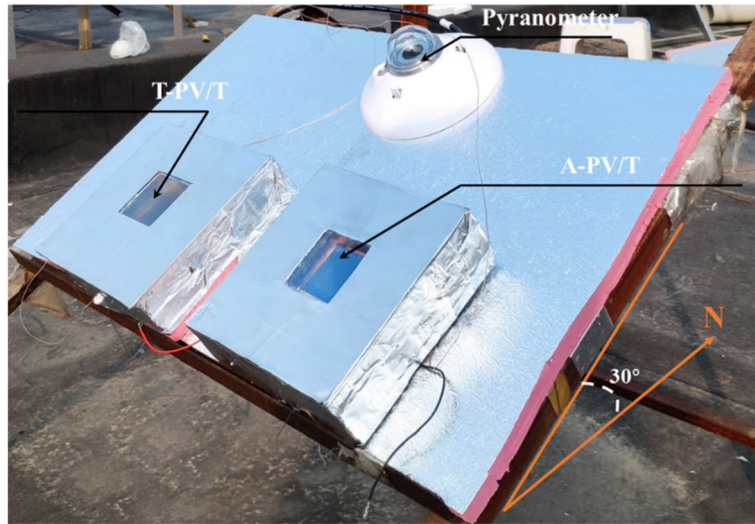


Fig. 6 Photo of the outdoor experiment setup

$$U_f = \sqrt{\sum_{i=1}^n \left(\frac{\partial f}{\partial x_i} \right)^2 (\Delta x_i)^2} \quad (2)$$

where U_f is the uncertainty of the parameter f , x_i and Δx_i are the i^{th} variable and its uncertainty.

3 Mathematical model

The energy flow process of the A-PV/T is presented in Fig. 7. Sunlight is transmitted through the silica aerogel and absorbed by the PV/T absorber with the generation of electricity and heat. Meanwhile, thermal loss occurs from the absorber to the environment. Generally, the energy control equation can be described by:

$$Q_{\text{th}} = Q_{\text{abs}} - Q_{\text{loss}} - Q_{\text{pv}} \quad (3)$$

where Q_{th} is the collected heat flux, Q_{abs} denotes the absorbed solar irradiance, Q_{loss} denotes the overall heat flux loss, and Q_{pv} represents the electrical power. The determination of the collected heat flux Q_{th} is related to the optical performance of the bulk aerogel. The PV/T total heat loss consists of two parts that include the heat dissipation power at the top surface (i.e., Q_{loss1}) and bottom surface (i.e., Q_{loss2}). The generated electricity can be calculated by applying an empirical correlation that involves the referenced electrical efficiency, PV cell temperature coefficient, and operating temperature. The detailed equations are described in [Appendix](#).

Thermal efficiency (η_{th}), electrical efficiency (η_{e}), and overall exergy efficiency (η_{ov}) are used as indicators to evaluate the performance of the A-PV/T, which are described by Eqs. (4), (5), and (6), respectively [33]:

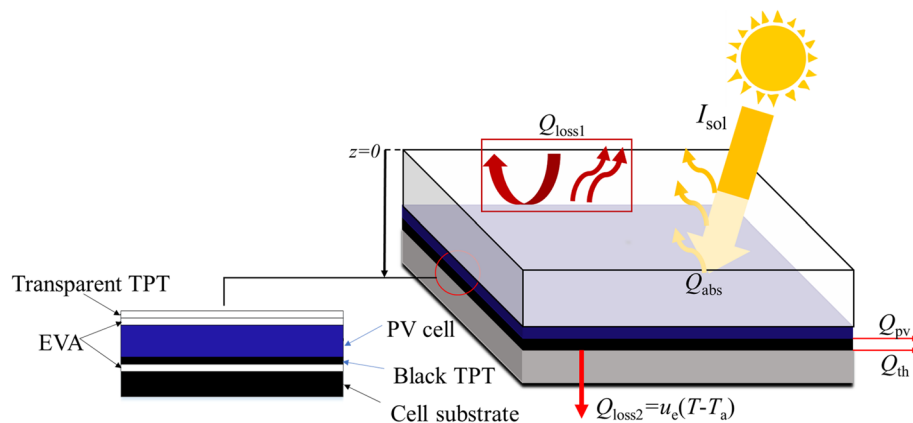


Fig. 7 The energy flow process of the A-PV/T

$$\eta_{th} = \frac{Q_{th}}{I_{sol}} \quad (4)$$

$$\eta_e = \frac{Q_{pv}}{I_{sol}} \quad (5)$$

$$\eta_{ov} = \frac{Q_{pv} + (1 - T_{amb}/T)Q_{th}}{E_{sol}} \quad (6)$$

where I_{sol} denotes the total solar irradiance, T and T_{amb} represent the PV/T absorber temperature and ambient air temperature, respectively. E_{sol} is the exergy of solar irradiance [34], which is related to sun temperature (T_{sun}), ambient temperature, and total solar irradiance:

$$E_{sol} = \left[1 - \frac{4}{3} \left(\frac{T_{amb}}{T_{sun}} \right) + \frac{1}{3} \left(\frac{T_{amb}}{T_{sun}} \right)^4 \right] I_{sol} \quad (7)$$

4 Results and discussions

Experiments demonstration of the A-PV/T and T-PV/T collectors are performed. Besides, the behavior of the two types of collectors under different conditions is further predicted and compared.

4.1 Thermal efficiency

Thermal efficiency of the A-PV/T and T-PV/T with different operating temperatures are measured using the system described in Fig. 4. Due to the difference in thermal conductivity between the glass and heat flux sensors, the value detected by the heat flux sensor is not the actual heat flux. Thus, calibration testing of the thermal efficiency testing system is conducted. The calibration testing system is illustrated in Fig. 8(a). A heater plate is placed on the surface of the a-Si cell and is thermally insulated to avert extra heat loss. Figure 8(b) presents the voltage output of the heat flux sensor under disparate heat power supplies, which shows that the system reaches

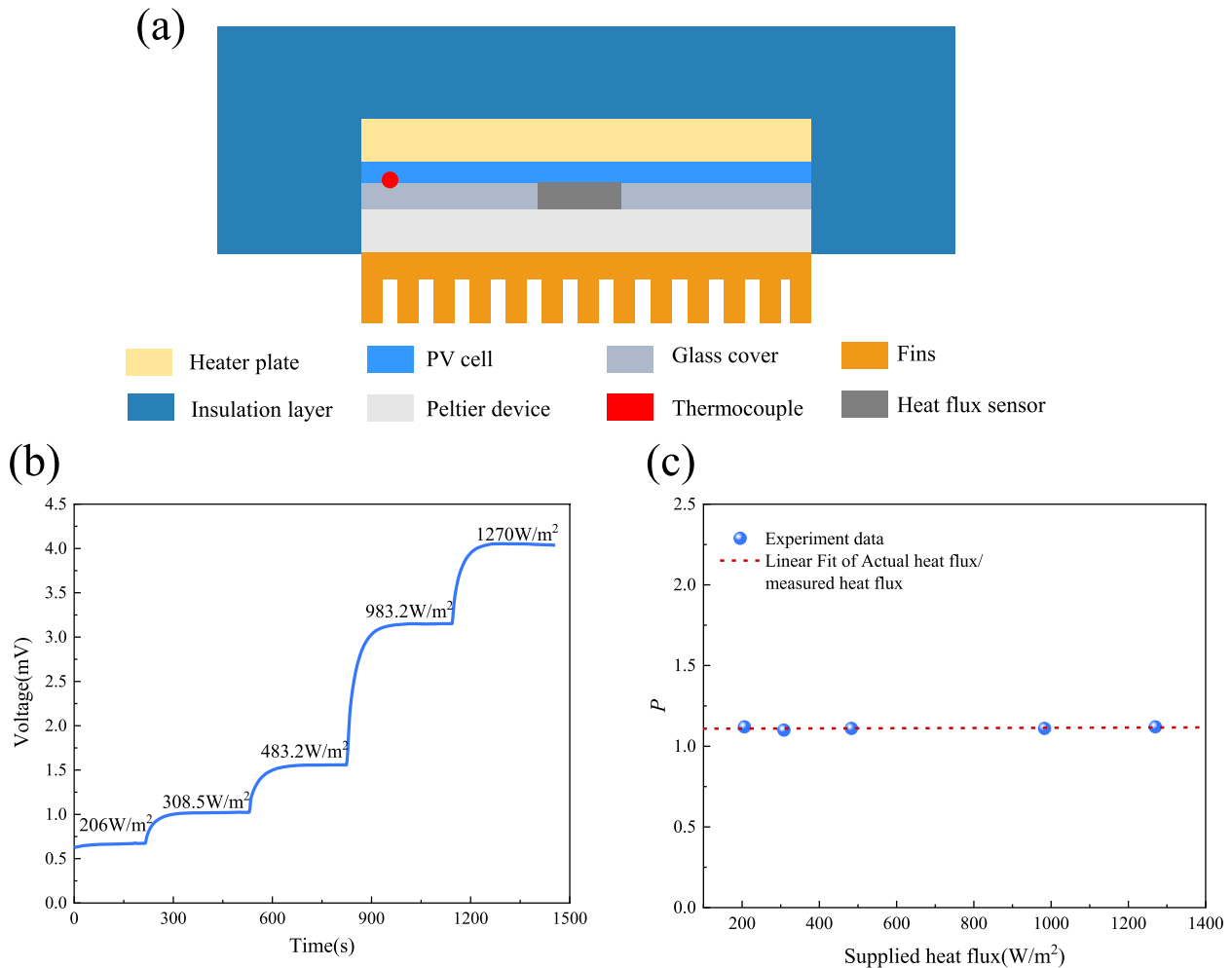


Fig. 8 **a** Calibration testing system. **b** Variation in the voltage of the heat flux sensor. **c** The ratio P is the ratio of actual heat flux to measured heat flux under different heating powers

a steady state quickly. As illustrated in Fig. 8(c), ratio P is used to eliminate the effect of the conductivity difference and obtain an accurate and steady heat flux value that has no connection with the supplied heating powers. The fitting result shows that the ratio P is approximately 1.11.

Figure 9 exhibits the variation in the two devices under different PV/T absorber temperatures. Compared with the T-PV/T, A-PV/T always exhibits higher thermal efficiency, regardless of the operating temperature. Because of the strong thermal loss suppression by aerogel, the A-PV/T expresses a better thermal performance than T-PV/T. The typical operating temperature range of previously reported PV/T collectors is approximately 35 °C (low solar irradiance [35])–70 °C (high ambient temperature and high solar irradiance [36, 37]), and the A-PV/T exhibits nearly 25.1% and 348% enhancement in thermal performance within this temperature range when compared with the T-PV/T. In addition, the T-PV/T possesses a stagnation temperature of 77 °C, which indicates that there is no potential for further heat collection. However, the A-PV/T can still possess a high thermal efficiency of 25.4% at 77 °C, which can provide a heat source for more applications. The main reason is that heat loss, especially radiative heat loss, increases rapidly when the surface temperature arises, and the thermal efficiency tends to zero after a rapid decline. In addition, the maximum uncertainties of thermal efficiency and electrical efficiency for A-PV/T are evaluated as 1.36% and 0.089%, respectively, and those for T-PV/T are evaluated as 1.28% and 0.10%, respectively.

4.2 Stagnation temperature, electrical efficiency, and overall exergy efficiency

To further explore the operating temperature region of the A-PV/T and T-PV/T, stagnation temperature and electrical efficiency are first measured under the solar simulator and the results are presented in Fig. 10. The temperature of the cell increases rapidly in the initial testing stage and eventually reaches a steady value. It is found that the temperature of the a-Si cell in the A-PV/T is always greater than that in T-PV/T. Specifically, the cell in the A-PV/T and T-PV/T reaches 103.1 °C and 77.3 °C after 20 min from the testing start, which indicates that the heat loss power of the T-PV/T is higher than that of the A-PV/T, showing a potential of aerogel for the efficiency improvement for solar energy harvesting.

It is found from Fig. 10(b) that the electrical efficiency of the a-Si cell with silica aerogel attached is relatively lower than the a-Si cell with a glass cover attached because the effective solar transmittance of the silica aerogel used in this study is slightly lower than that of the glass, which means that there exists an extra optical loss in the T-PV/T collector. Notably, the solar transmittance of the silica aerogel is being improved as manufacturing techniques developed. At present, solar transmittance of the silica aerogel is improved to 0.95 and this is experimentally proven and reported in ref. [27]. Also, it is found that the electrical efficiency drops slightly with increasing temperature because the solar cell has a negative temperature coefficient. However, the a-Si cells used in this study possess a low-temperature coefficient with

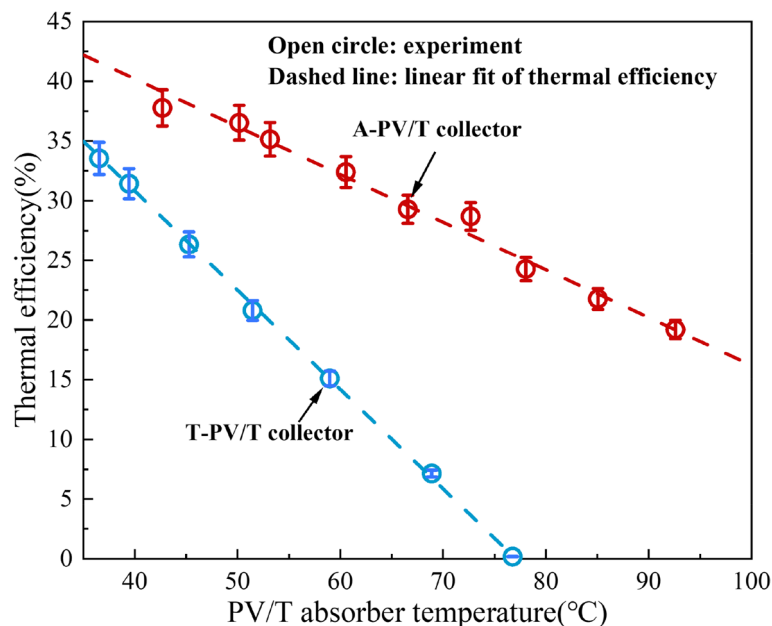


Fig. 9 Measured and fitted thermal efficiency of the A-PV/T and the T-PV/T

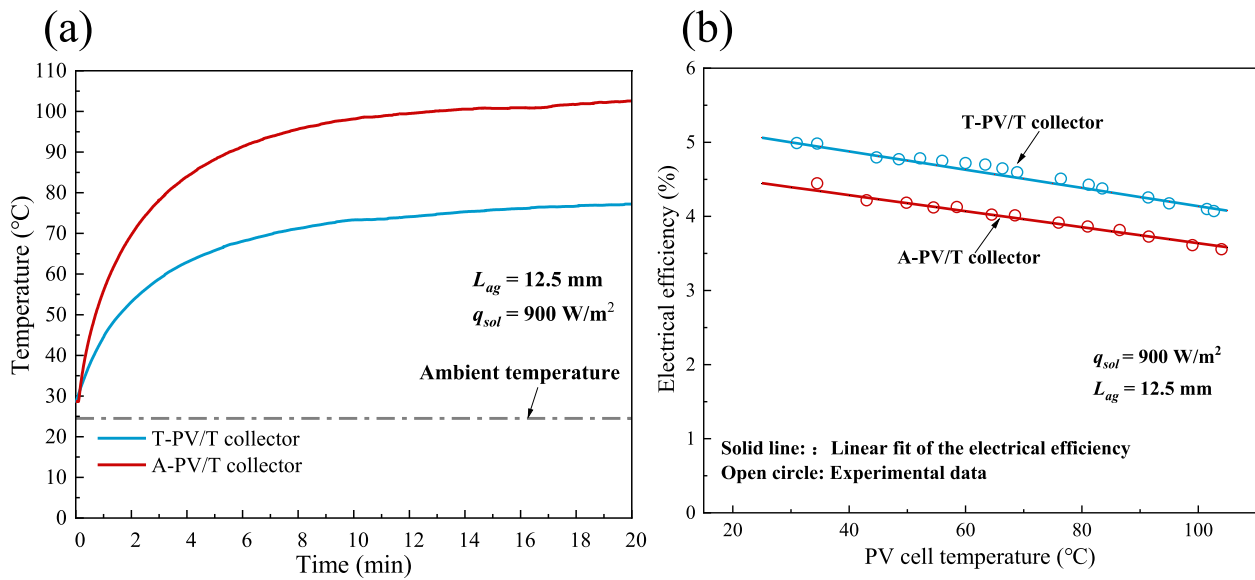


Fig. 10 Temperature and electrical efficiency of the a-Si cell. **a** Temperature of the a-Si cell in the A-PV/T and T-PV/T. **b** Electrical efficiency of the a-Si cell with silica aerogel and glass at different temperatures

a value of $-0.24\%/^{\circ}\text{C}$, which means that the sensitivity of the a-Si cell to temperature is lower than that of widely used Si cells. Notably, the temperature coefficient of PV cells could be theoretically positive and was successfully reported using the InGaN solar cell [38], which provides a new pathway for PV/T applications.

As shown in Fig. 11, compared with T-PV/T collectors, A-PV/T collectors can obtain a higher overall exergy efficiency when the operating temperature

exceeds 45°C . The exergy efficiency of A-PV/T is lower than that of T-PV/T when the working temperature was lower due to optical loss caused by aerogel. At an operating temperature of 60°C , the exergy efficiency of the A-PV/T collector is 1.21 times that of the T-PV/T collector. This ratio increases to 1.45 as operating temperatures rise to 70°C . It is worth noting that the best operating temperature of A-PV/T is 74°C with an exergy efficiency of 8.29%. Compared with the T-PV/T

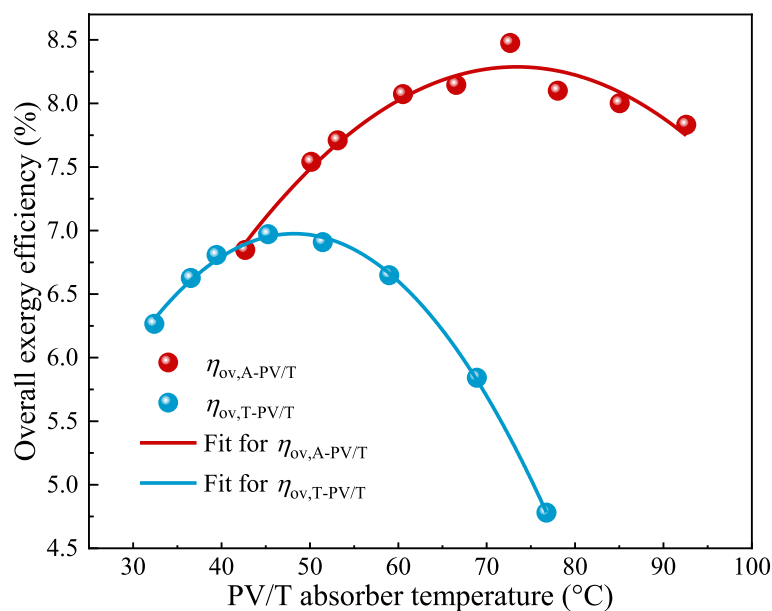


Fig. 11 Overall exergy efficiency of the A-PV/T and T-PV/T at different temperatures

collector (the optimal working temperature of 48 °C, and the maximum exergy efficiency of exergy was 6.97%), A-PV/T has a higher optimal working temperature and a greater exergy efficiency, which is conducive to the application of PV/T technology in more scenes. Importantly, when the solar transmittance of the silica aerogel is increased to the level of the glass, the exergy efficiency will be higher than that of the T-PV/T at all working temperatures, and this is what needs to do in the future.

Outdoor experimental testing was also performed and constructed on November 12th, 2020, from 9:00 to 15:00 to further explore the behavior of the A-PV/T in actual sky conditions. Meteorological parameters, including ambient air temperature and solar irradiance, are presented in Fig. 12(a). Figure 12(b) describes the variations in the a-Si cell temperature under different PV/T collectors. The trend of the cell temperature in the two PV/T collectors is highly correlated with the variation in solar irradiance. It can be observed that the A-PV/T collector always presents supernal temperatures than the T-PV/T. For instance, the solar cell in the A-PV/T and T-PV/T reach maximum temperatures of 96.7 °C and 70.7 °C, respectively. In addition, the maximal temperature difference reaches 27.4 °C. The high stagnation temperature of the A-PV/T benefits from the ability of the aerogel to reduce heat loss. The same results are obtained in both indoor and outdoor experiments, which demonstrate the feasibility of the PV/T collector combined with aerogel. Notably, such a high stagnation temperature does not mean that the PV cell will reach this temperature in actual work. In actual work, the cell temperature can be controlled by the operation condition of the working

fluid, such as using highly effective nano-fluid or improving flow rates of the working fluid.

4.3 Numerical analysis

Distinct performance of the PV/T utilization in different regions occurs due to distinct weather conditions. To clarify the annual performance behavior in different regions, a theoretical model is carried out for simulation study.

4.3.1 Model validation

Model validation is conducted with the practical data obtained from the experimental setup. The meteorological boundary parameters are from the experiment in Sect. 4.1. As plotted in Fig. 13, the predicted thermal efficiency is consistent with the experimental results, which shows the simulation and experimental results present a similar trend and the RMSD is 3.81% over the whole temperature range. This tiny distinction shows that this model is credible for the performance prediction in this study.

4.3.2 Performance in different regions

The annual performance of the A-PV/T and T-PV/T in Beijing, Fuzhou, Urumqi, and Hefei are explored by the mathematical model. The thermal and electrical energy gains are presented in Figs. 14 and 15. During the simulation, a 20-mm-thick aerogel is applied to the collector, the operating temperature is set as 60 °C, the PV/T absorber exhibits an emissivity of 0.95, and the areas of the two types of collectors are assumed to be $1 \times 1 \text{ m}^2$, the daily operation time of Beijing, Fuzhou, Hefei is set from 8:00 to 16:00, while the operation time of Urumqi

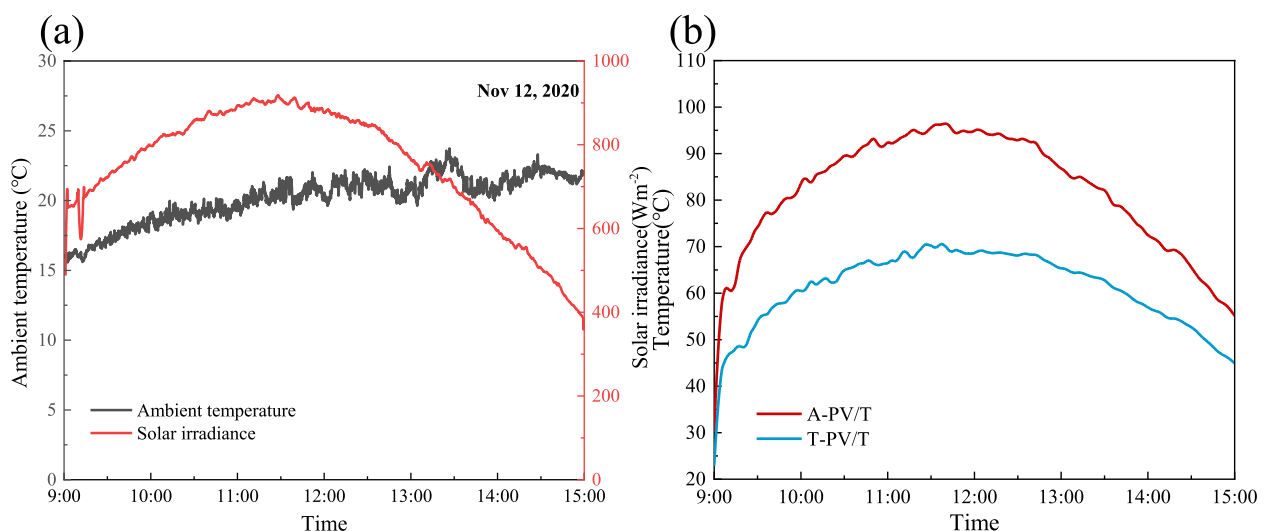


Fig. 12 **a** Measured ambient temperature and solar irradiance. **b** Measured cell temperature in two PV/T collectors

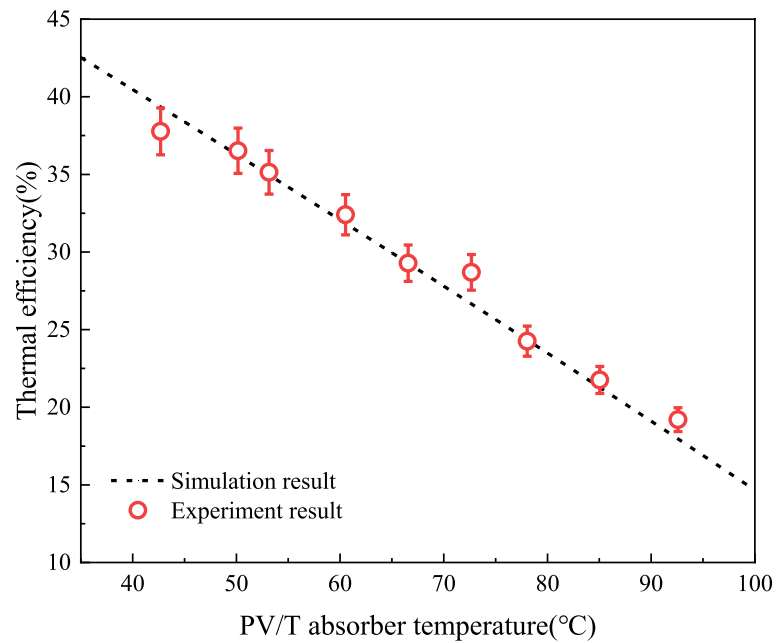


Fig. 13 Numerical and experimental data of the thermal efficiency

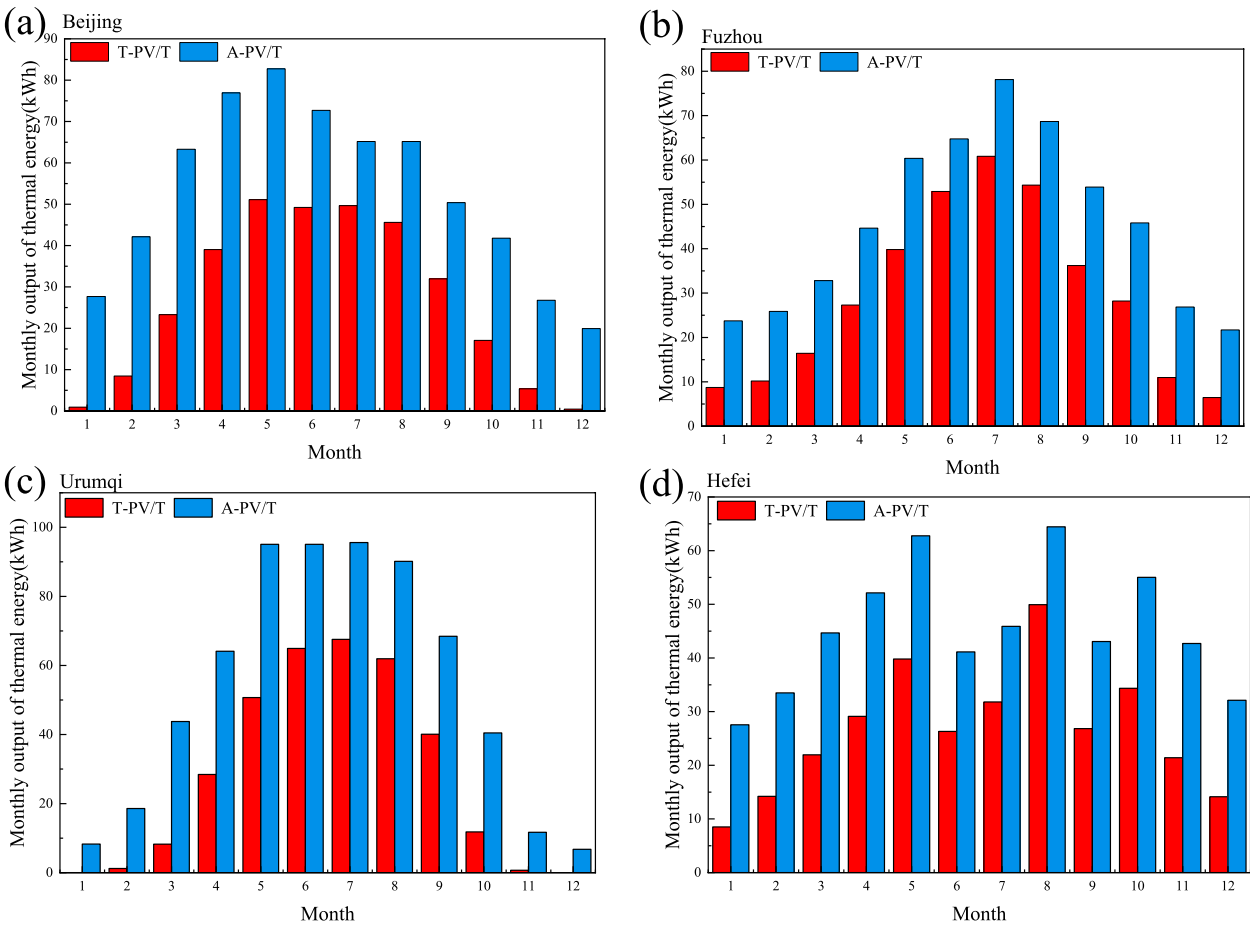


Fig. 14 Monthly output of thermal energy in a Beijing, b Fuzhou, c Urumqi, and d Hefei

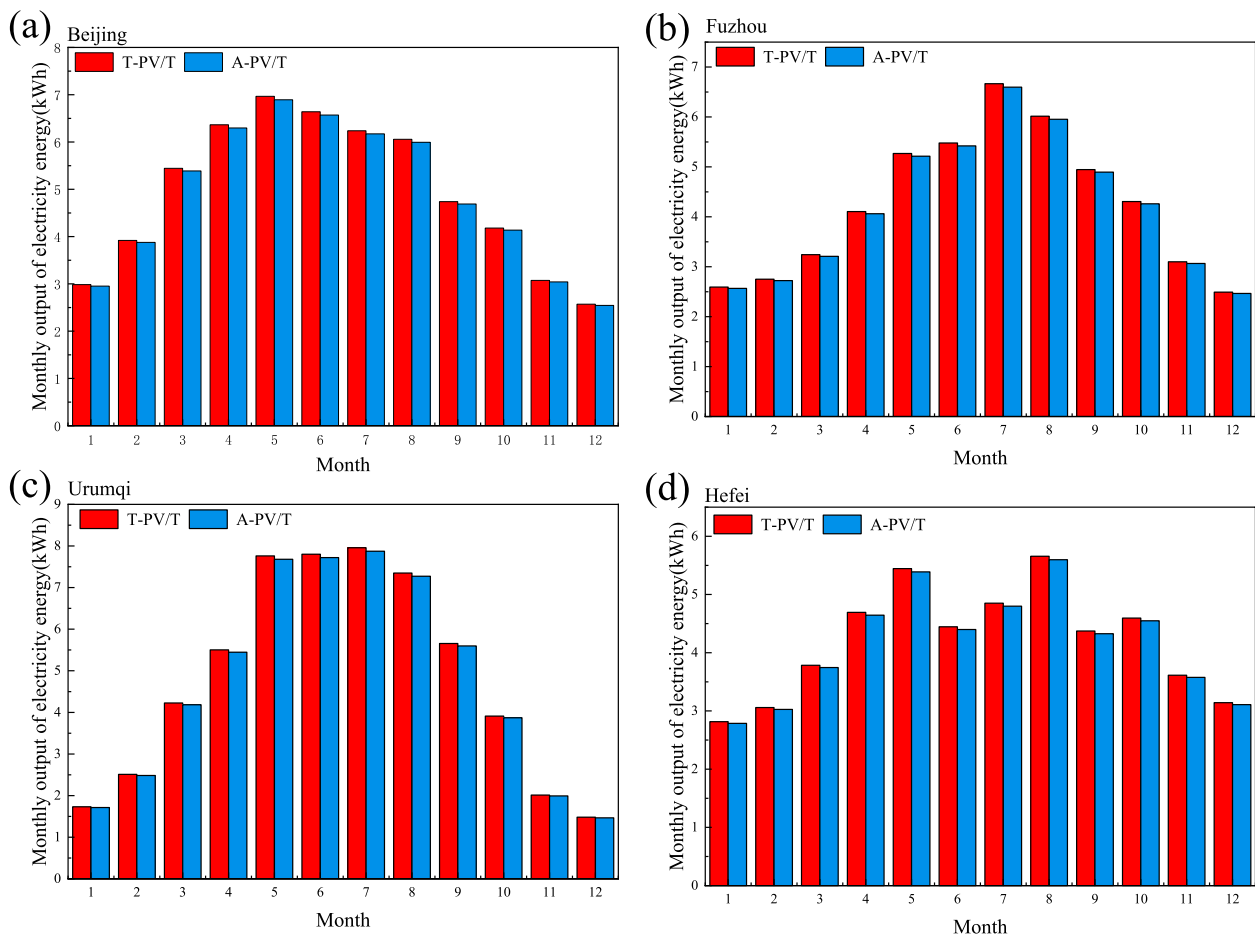


Fig. 15 Monthly output of electricity in **a** Beijing, **b** Fuzhou, **c** Urumqi, and **d** Hefei

is located from 10:00 to 18:00. All weather condition data are obtained from EnergyPlus [39].

As displayed in Fig. 14, the A-PV/T possesses a higher monthly heat gain than that of the T-PV/T, especially in cold seasons. For example, in Urumqi and Beijing, the T-PV/T cannot work properly in winter due to the ultralow ambient temperature. In January, the A-PV/T can provide 27.7 kWh and 8.3 kWh thermal energy gain in Beijing and Urumqi, respectively, while the T-PV/T produces only 0.9 kWh and 0.027 kWh, respectively. The annual thermal energy gain and electricity gain of the four regions are summarized in Table 1, which shows the A-PV/T in Beijing, Fuzhou, Urumqi, and Hefei can obtain total thermal energy of 638.7 kWh, 547.3 kWh, 634.1 kWh, and 545.0 kWh, respectively, which are 1.9, 1.5, 1.8, and 1.7 times those of the T-PV/T, respectively. The great thermal performance of A-PV/T is benefit from thermal loss suppression of the aerogel. Specifically, during the month when the heat demand is greatest, the A-PV/T can provide thermal energy continuously. These predicted results indicate that A-PV/T is more suitable

Table 1 Annual performance comparison of A-PV/T and T-PV/T in Beijing, Fuzhou, Urumqi, and Hefei

Region	T-PV/T		A-PV/T	
	Thermal energy gains (kWh)	Electricity gains (kWh)	Thermal energy gains (kWh)	Electricity gains (kWh)
Beijing	322.1	59.2	638.7	58.6
Fuzhou	352.3	51.0	547.3	50.4
Urumqi	335.7	57.9	634.1	57.3
Hefei	318.4	50.5	545.0	49.9

for year-round operation than T-PV/T, especially in cold regions.

As presented in Fig. 15, the monthly electrical energy gains of the A-PV/T are slightly lower. The annual electrical energy gains of the A-PV/T in Beijing, Fuzhou, Urumqi, and Hefei are 58.6 kWh, 50.4 kWh, 57.3 kWh, and 49.9 kWh, respectively, which are 1.01%, 1.18%, 1.04%, and 1.19% lower than those of the T-PV/T

collector, respectively. The electrical reduction is caused by the optical loss of the aerogel. However, with the optimization of the manufacturing process, the silica aerogel can be fabricated with a high transmissivity and low optical loss for solar irradiance, and this viewpoint was experimentally proven [27]. Although the electrical energy gains of the A-PV/T decreased throughout the year, the decrease in electrical performance was negligible compared to the enhancement in thermal energy gains. Most importantly, in the season when heat is most needed in cold regions, the T-PV/T collector can hardly work normally, but the A-PV/T still has a high heat output, which makes up for the lack of seasonal inability to supply useful heat for end-use (e.g., space heating).

5 Conclusions

In summary, an innovative A-PV/T was proposed and investigated based on the feature that silica aerogel with ultra-low thermal conductivity is solar transparent and thermally insulated. Indoor and outdoor experimental tests were conducted for performance evaluation. In addition, a mathematical model is developed and validated for performance prediction in different regions. Some main conclusions are summarized as follows:

- (1) The thermal efficiency of the A-PV/T collector is 25.1%–348% greater than that of the T-PV/T collector when the collection temperature is within the typical operating temperature range of 35 °C–70 °C, indicating that silica aerogel can significantly improve its thermal efficiency.
- (2) The stagnation temperature of the A-PV/T and T-PV/T can reach 103.1 °C and 77.3 °C, respectively, which shows that silica aerogel can significantly suppress the radiative heat loss of the PV/T collector.
- (3) Annual thermal energy gains of the A-PV/T collector in Beijing, Fuzhou, Urumqi, and Hefei are 638.7 kWh, 547.3 kWh, 634.1 kWh, and 545.0 kWh, respectively, which are 1.9, 1.5, 1.8, and 1.7 times those of the T-PV/T.
- (4) The A-PV/T improves the seasonal suitability to provide useful heat in cold seasons for hot water supply and space heating when T-PV/T cannot efficiently provide the required thermal energy.

In the future, more efforts should be paid to improve the performance of the silica aerogel and corresponding PV/T collectors, such as enlarging the solar transmittance of the silica aerogel, improving the mechanical strength of the aerogel, and achieving cost-effective large-scale manufacture.

Appendix

Theoretical model of the A-PV/T

As shown in Fig. 7, the detailed mathematical model is as follows. The collected heat flux is expressed as:

$$Q_{th} = Q_{abs} - Q_{loss} - Q_{pv} \quad (8)$$

Q_{abs} and Q_{pv} are expressed as Eqs. (9) and (10), respectively:

$$Q_{abs} = \tau_a \tau_{ad} I_{sol} \alpha \quad (9)$$

$$Q_{pv} = I_{sol} \tau_a \tau_{ad} \eta_{e,ref} [1 + B_{pv}(T - T_{ref})] \quad (10)$$

where τ_a and τ_{ad} are the solar transmittances of the aerogel and adhesive, I_{sol} is total solar irradiance, α is the effective solar absorption, B_{pv} denotes the power-based temperature coefficient of the cell, and $\eta_{e,ref}$ is the referenced PV efficiency ($T_{ref} = 25^\circ\text{C}$).

The radiative transfer equation (RTE) is used and coupled with the heat conduction governing equation to calculate the heat flux within the aerogel. The RTE is expressed as [40]:

$$\frac{dI_\lambda(z, \vec{s})}{dz} = \kappa_\lambda I_{b\lambda}(z) - (\sigma_\lambda + \kappa_\lambda) I_\lambda(z, \vec{s}) + \frac{\sigma_\lambda}{4\pi} \int_{4\pi} I'_\lambda(z, \vec{s}') \Phi_\lambda(\vec{s}, \vec{s}') d\Omega' \quad (11)$$

$$k \frac{d^2 T}{dz^2} - \frac{dq_{\lambda=0 \sim \infty}^r}{dz} = 0 \quad (12)$$

where I_λ denotes the radiative intensity field, σ_λ and κ_λ denote scattering and absorption coefficient, $\Phi_\lambda(\vec{s}, \vec{s}')$ denotes the phase function, $I_{b\lambda}$ denotes the blackbody intensity, k denotes the solid thermal conductivity of the silica aerogel, z is the spatial coordinate along with the thickness of the aerogel, T denotes the temperature field within the aerogel, and $q_{\lambda=0 \sim \infty}^r$ denotes the spatial radiative heat flux. When $T(z)$ has been determined, the radiative and conductive heat flux can be then obtained at any z within the aerogel. Therefore, thermal loss of the PV/T absorber to the environment through the silica aerogel can be obtained by summing the radiative and conductive heat flux at the top surface of the aerogel ($z = 0$) [41].

$$\begin{cases} Q_{loss1} = q_{\lambda=0 \sim \infty}^r(z=0) + q_{cond}(z=0) \\ q_{\lambda=0 \sim \infty}^r(z=0) = \int_0^\infty \int_{4\pi} I_\lambda(\vec{s}) \vec{s} d\Omega d\lambda \\ q_{cond}(z=0) = -k \frac{dT}{dz} \Big|_{z=0} \end{cases} \quad (13)$$

where $q_{\lambda=0 \sim \infty}^r(z=0)$ is the spatial radiative heat flux obtained by RTE at the top surface of the aerogel, $q_{cond}(z=0)$ is the conductive heat flux at the top surface of the aerogel.

Another part of the thermal loss is transferred from the PV/T absorber to the environment through the back-insulation layer, which is expressed by:

$$Q_{\text{loss}2} = u_e \cdot (T - T_{\text{amb}}) \quad (14)$$

where T_{amb} is ambient temperature, u_e is the thermal loss coefficient between the PV/T absorber and environment, which can be determined by:

$$u_e = \frac{1}{L/k + 1/h} \quad (15)$$

where L is the thickness of the insulation layer. h is the convection loss coefficient between the insulation layer and the environment and can be estimated by $h = 2.8 + 3.0 V$, where V is wind speed.

The stagnation temperature is obtained when the thermal efficiency is 0, which is the maximum achievable temperature collector [42]. The PV/T collector can supply useful thermal energy when the operating temperature is lower than the stagnation temperature.

The mathematical model is set up according to the aforementioned mathematical equations. The root mean square deviation (RMSD) is applied to validate the predicted results [43]:

$$RMSD = \sqrt{\frac{\sum [(T_{\text{sim},i} - T_{\text{exp},i})/T_{\text{exp},i}]^2}{n}} \quad (16)$$

Abbreviations

PV	Photovoltaic
PV/T	Photovoltaic/thermal
RTE	Radiative transfer equation
RMSD	Root-mean-square deviation
TPT	Tedlar-Polyester-Tedlar
A-PV/T	Aerogel photovoltaic/thermal
T-PV/T	Traditional photovoltaic/thermal
AM	Air mass

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Authors' contributions

WL proposed the design and performed the testing and simulation. GD, JD, and HM contributed to the analysis and paper review. PG and ZB conceived the study and edit the manuscript. All authors read and approved the final manuscript.

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Availability of data and material

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

Gang Pei is an editorial board member for Carbon Neutrality and was not involved in the editorial review, and the decision to publish this article. All authors declare that there are no competing interests.

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